



# SPS Research Framework

Supervisory Prioritisation Score for Retail Payment Systems

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Challenge 2026 – NayaOne Sandbox

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DORA EU 2022/2554

PSD2 EU 2015/2366

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# Notebook Structure

Eight sequential phases transform raw transaction data into actionable supervisory intelligence.

01

## Data Ingestion & Topological Analysis

GLM instrument vulnerability deep dive

02

## AutoML Genetic Optimiser

Calibrate  $\alpha$  (technical) and  $\beta$  (social) weights

03

## SPS Recalculation & Multi-Algorithm ML

XGBoost, Random Forest, Logistic Regression

04

## Dashboard & Vigilance Matrix

4-class PSP segmentation and validation

05

## Network Topology + Final Validation

DORA stress test & PSD2 heatmap

06

## Time-Series Surveillance

ARIMA (744 h) + SARIMA (12 months)

# Phase 1 – Data Ingestion & Fusion

## Dataset

**414K**

Transactions

**698**

PSPs

**20**

Columns

## Data Sources (NayaOne API)

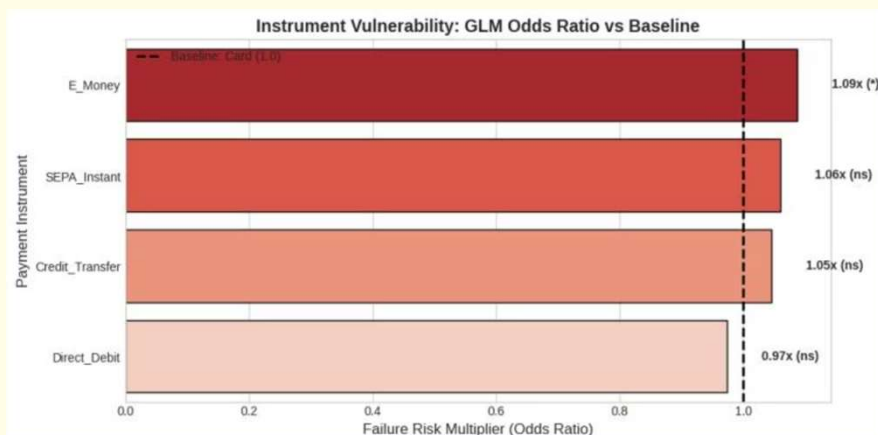
- D01 Transactions · D06 PSP Registry
- D03 Performance · D07 Channels · D10 Profiles

## Missing Data Handling

34.19% of `processing_time_ms` missing (Completed transactions only). Imputed via per-instrument median – Card: 1,479 ms · Credit Transfer: 5,412,169 ms · Direct Debit: 5,420,446 ms.

# Phase 1.5 – GLM Instrument Vulnerability

## Logistic Regression vs Card Baseline



E\_Money is the only instrument with a statistically significant higher failure risk (OR = 1.087,  $p = 0.040$ ). SEPA\_Instant and Credit\_Transfer show marginal elevation but are non-significant. Direct\_Debit is marginally below baseline.

## Real Failure Rates

Global mean: 2.02%. E\_Money leads at 2.16%, Direct\_Debit lowest at 1.94% – a narrow band confirming overall system stability.

## Phase 2 – AutoML Genetic Optimiser



Convergence reached at generation 10 – fitness stable for 10 consecutive generations (std = 0.000000).

### Optimisation Results

#### $\alpha$ (Technical Weight)

0.2404 – DORA operational resilience

#### $\beta$ (Social Weight)

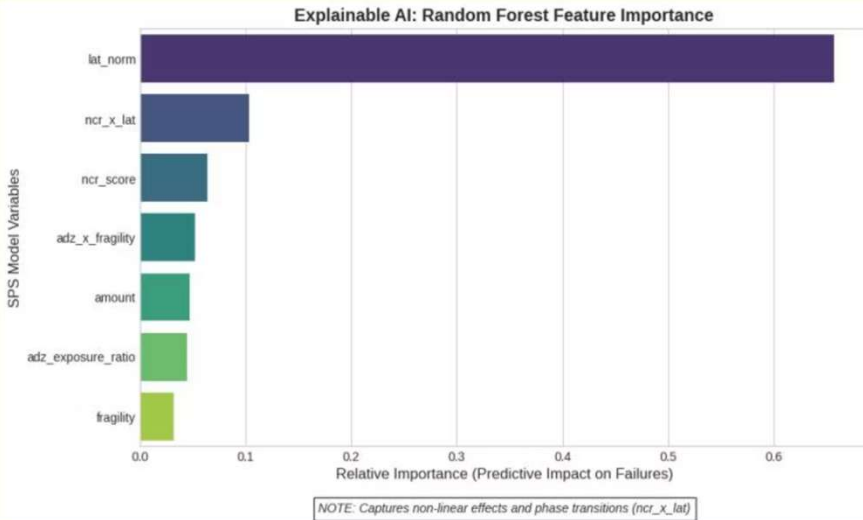
0.2404 – PSD2 financial inclusion

#### Ratio $\alpha/\beta$

1.00:1 – Balanced bi-focal risk

☐ **Vigilance Diagnosis:** Equal weights confirm prevalence of **technical risks** (hub failure). Priority: DORA inspections on high-NCR hubs.

# Phase 3 – Machine Learning Inference



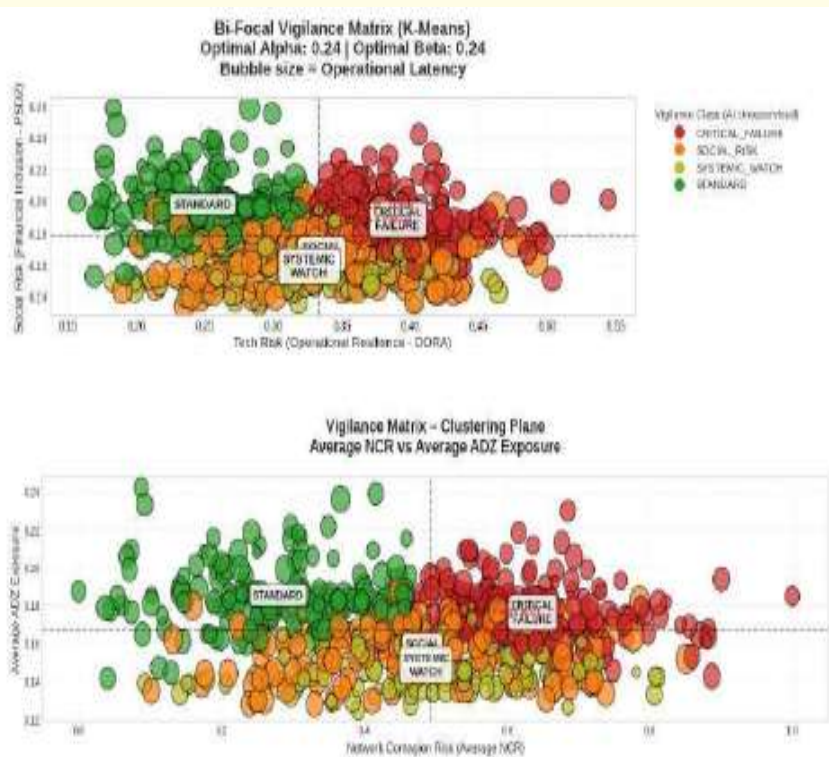
## Multi-Algorithm Comparison

| Algorithm           | ROC-AUC     | F1     |
|---------------------|-------------|--------|
| XGBoost             | 0.6855<br>✓ | 0.0618 |
| Random Forest       | 0.6845      | 0.0599 |
| Logistic Regression | 0.5038      | 0.0000 |

## Explainable AI – Top Features

- **lat\_norm**: 65.7% – operational congestion is the main driver
- **ncr\_x\_lat**: 10.4% – confirms NCR × Latency phase transitions
- **ncr\_score**: 6.4% – network centrality matters

Non-linearities captured by XGBoost validate the evolutionary SPS approach. PR-AUC is the most informative metric for this imbalanced use-case (2.02% minority class).



## Phase 4 – Vigilance Matrix

### ● Critical Failure

201 PSPs (28.8%) SPS mean: 0.1393 · NCR mean: 0.6356  
→ Immediate DORA inspection

### ● Social Risk

167 PSPs (23.9%) SPS mean: 0.1206 · Social: 0.1666  
→ PSD2 inclusion audit

### ● Systemic Watch

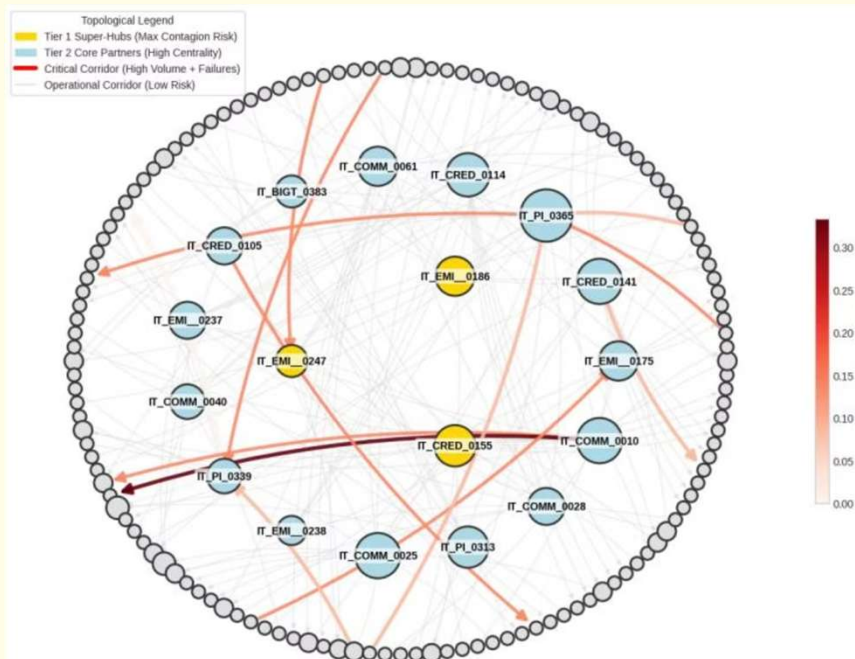
159 PSPs (22.8%) SPS mean: 0.1166  
→ Continuous monitoring

### ● Standard

171 PSPs (24.5%) SPS mean: 0.1073 · NCR mean: 0.2792  
→ No action required

K-Means clustering is validated: SPS ranking follows the expected hierarchy **CRITICAL > SOCIAL > SYSTEMIC > STANDARD** with a 0.0320-point spread between extremes.

# Phase 5 – Network Topology



## Scale-Free Topology

Power-law exponent  $\Gamma = 2.7892$ , aligned with the Barabási-Albert model – confirming the network is scale-free and vulnerable to targeted hub removal.

## Tier 1 Super-Hubs (Max Betweenness)

### DORA Stress Test

All three scenarios (single, triple, systemic shock) returned **Fragmentation Index = 0.0** and **Blackout = 0%** – network shows good topological resilience.  
Recovery Complexity: 5.

**IT\_EMI\_\_0186**

NCR = 1.0000

**IT\_EMI\_\_0247**

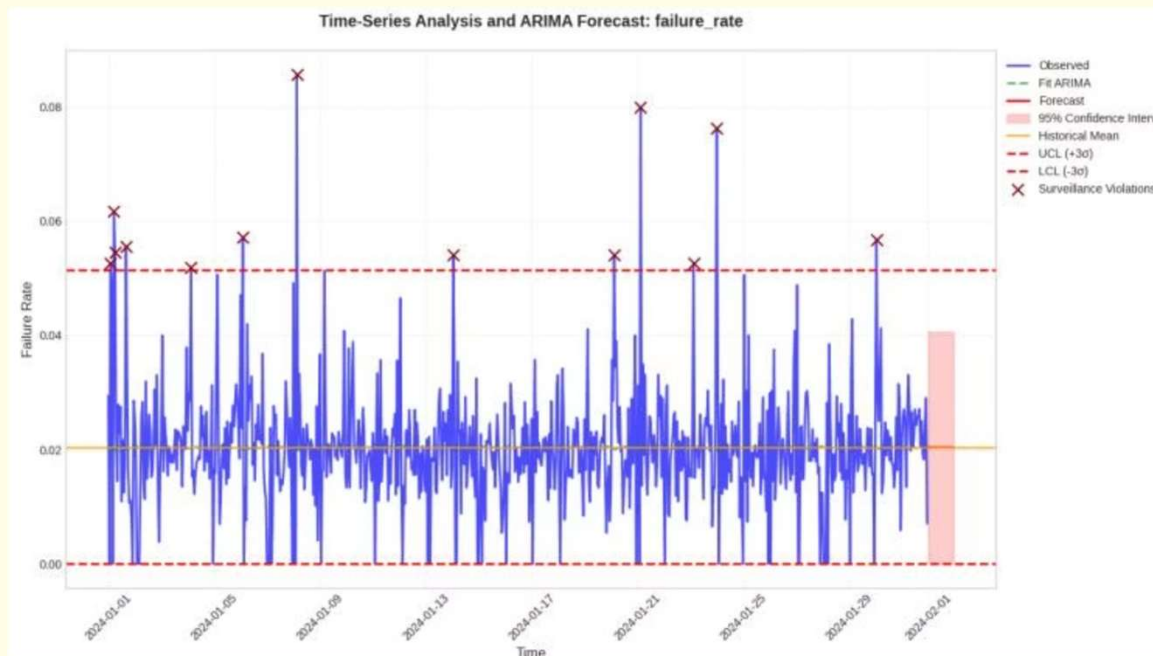
NCR = 0.9002

**IT\_CRED\_0155**

NCR = 0.8867



# Phase 7 & 8 – Time-Series Surveillance



## ARIMA – Hourly (Phase 7)

- 744 hourly periods analysed (Jan 2024)
- All three series stationary (ADF  $p = 0.000$ )
- ARIMA(2,0,2) fitted for failure\_rate, avg\_latency, avg\_ncr
- Surveillance violations flagged where data exceeds UCL ( $3\sigma$ )

## SARIMA – Monthly (Phase 8)

- 12 months of data, ~400K–430K transactions/month
- Coefficient of Variation <10% for SPS and NCR  
→ STABLE
- SARIMA confirms absence of critical seasonal trends



## Key Results & Supervisory Recommendations



### DORA Priority

201 PSPs (28.8%) require immediate inspection – focus on latency and hub congestion



### PSD2 Priority

167 PSPs (23.9%) require inclusion audit – focus on Active Denial Zones (ADZ)



### Scale-Free Network

$\Gamma = 2.7892$  · 3 Tier-1 super-hubs identified · resilience confirmed under stress



### Predictive ML

XGBoost ROC-AUC = 0.6855 · lat\_norm dominates (65.7%) · non-linearities validated



**Bi-focality validated:** Both DORA (operational resilience) and PSD2 (financial inclusion) risks are material. The SPS framework is ready for submission to the Innovation Data Challenge 2026 and presentation to supervisory authorities.